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B. E. (Third Semester) Examination, Nov.-Dec. 2016

(New Scheme)

(Electrical Branch)

ELECTRICAL MACHINES-I

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note *: Attempt all questions. Part (a) is compulsory from each questions. Attempt any two parts from (b), (c) and (d).*

1. (a) What is transformation ratio? Why transformer rating in kVA? 2

(b) Write short notes on any two : 7

(i) Efficiency curve of transformer

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(ii) OCT & SCT in transformer

(iii) Condition of zero and maximum regulation in transformer.

(c) Draw the phasor diagram of single phase transformer at different power factor of load. 7

(d) Derive that the per unit voltage regulation of a transformer is given by $V_{sc}\cos(\theta_{sc}-\theta_2)/E_2$ where V_{sc} is the voltage required to circulate rated current on short-circuit, θ_{sc} is the p.f angle on short circuit and θ_2 is the load power factor angle. 7

2. (a) State why distribution transformers are designed for lower iron losses. 2

(b) Write short notes on any two : 7

(i) Switching transients in transformer

(ii) Back to back test of transformer

(iii) All-day-efficiency for a transformer.

(c) Draw the phasor diagram and equivalent circuit of auto-transformer. 7

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(d) A 2300/230 V, 500 kVA, 50 Hz distribution transformer has core loss of 1600 W at rated voltage and copper loss 7·5 kW at full load. During the day it is loaded as follows :

% Load	0%	20%	50%	80%	100%	125%
Power Factor		0·7 lag	0·8 lag	0·9 lag	1	0·85 lag
Hours	2	4	4	5	7	2

Determine the all-day efficiency of the transformer. 7

3. (a) Enlist application of scott connected transformer. 2

(b) Write short notes on any two : 7

(i) Different vector groups

(ii) Open-delta connection

(iii) Three-winding transformer

(c) In open delta or v-v connection of transformer show : 7

(i) The kVA rating of open delta is 57·7% of closed delta rating.

(ii) Utilization factor is 86·6%.

(d) Draw the equivalent circuit of three winding

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transformer. How the parameter of equivalent circuit determined? 7

4. (a) What is meant by magnetic neutral plane (M.N.P.) 2

(b) Explain the construction of dc machine and derive the expression of generated emf. 7

(c) Explain 'Armature Reaction' in DC machine. Explain how the effect of armature reaction can be reduced? 7

(d) The magnetizing characteristics of DC shunt generator is given at 1000 rpm. 7

Field Current I	Gen. EMF E
1	160
2	260
4	390
6	472
8	522
10	550

Field resistance = 60 ohm, armature resistance is 0·5 ohm, load resistance = 3 ohm. Find the terminal voltage and load current.

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5. (a) Enlist various methods of speed control used for DC shunt motor. 2

(b) Explain the various characteristics of dc motor. 7

(c) Describe the operation of a 3-point starter. 7

(d) A belt driven 100 kW shunt generator running at 300 rpm on 220 V bus - bars continues to run as a motor when the belt breaks, then taking 10 kW. What will be its speed? Given armature resistance = 0.025 ohm. Field resistance = 60 ohm and contact drop under each brush = 1 V. Ignore armature reaction. 7

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